

Experiment 3: Information Theory

Aim

To calculate the entropy and information gain of all input attributes in the Play Tennis Dataset and identify the attribute with the highest information gain.

Program

```
import pandas as pd

from math import log2

data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain', 'Overcast',
               'Sunny', 'Sunny', 'Rain', 'Sunny', 'Overcast', 'Overcast', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Cool',
                   'Mild', 'Cool', 'Mild', 'Mild', 'Mild', 'Hot', 'Mild'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'Normal',
                'High', 'Normal', 'Normal', 'Normal', 'High', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak', 'Strong', 'Strong',
            'Weak', 'Weak', 'Weak', 'Strong', 'Strong', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes',
            'No', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'No']
}

df = pd.DataFrame(data)

def entropy(target):
    values = target.value_counts(normalize=True)
    return -sum(p * log2(p) for p in values)

def information_gain(df, attribute, target='Play'):
    total_entropy = entropy(df[target])
    weighted_entropy = 0
    for value in df[attribute].unique():
```

```

subset = df[df[attribute] == value]

weighted_entropy += (len(subset)/len(df)) * entropy(subset[target])

return total_entropy - weighted_entropy

print("Entropy of Dataset:", entropy(df['Play']))

print("\nInformation Gain")

best_attribute = None

best_gain = -1

for col in df.columns[:-1]:

    gain = information_gain(df, col)

    print(col, ":", round(gain,4))

    if gain > best_gain:

        best_gain = gain

        best_attribute = col

print("\nBest Attribute:", best_attribute)

print("Highest Information Gain:", round(best_gain,4))

```

Output



```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\personal\cloud computing\123.py =====
Entropy of Dataset: 0.9402859586706311

Information Gain
Outlook : 0.2467
Temperature : 0.0292
Humidity : 0.1518
Wind : 0.0481

Best Attribute: Outlook
Highest Information Gain: 0.2467
>>> |

```

Ln: 15 Col: 6

Result

The entropy and information gain were calculated successfully. Among all the attributes, Outlook had the highest information gain and was selected as the best attribute for splitting the dataset.